

MANE IN INDIA

We come out on top as scientists put human hair from three major ethnic groups to the test

Thickness

The diameter of Indian hair is greater than the other two types. It is 35% thicker than Caucasian hair and 25% thicker than African hair.

Strength

Tensile strength determines how easy it is to break hair. Indian hair has been rated the toughest while African hair is the weakest.

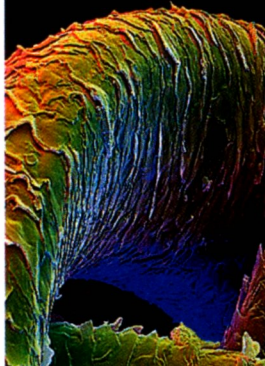
The World's Best Hair

Hair is now a subject of intense research. It used to be thought there are no major biochemical differences between the hair of various ethnic groups. A study conducted by TRI/Princeton, an independent, non-profit research institute based in New Jersey, US, and shared exclusively with INDIA TODAY, challenges that conventional wisdom. Led by Dr Yash K. Kamath, an eminent scientist in the field of fibre science, the study has found that multiple factors, from cultural practices to genetic and geographic variations, play a role in creating differences. What's more, Indian hair was found to have the best structural and mechanical characteristics compared to Caucasian and African hair.



TRESS CULTURE

15 hair samples from women in the age group of 25-35, from each of the three ethnic categories—African, Caucasian and Indian—were collected. Scanning electron microscope analysis of hair knots revealed that African hair was structurally more damaged, with uplifted cuticles and frayed hair-shafts compared to Caucasian hair.



African Excessive braiding and use of straighteners. Most damaged with a porosity of 47%.



Caucasian Blow drying, use of styling products, colouring give it a porosity of 47%.



Indian Regular oiling made it smooth. Least damaged with 35% of porosity.

Shine

The lustre of the hair depends on the level of damage to the strands. Indian oiled hair is 1.5 times shinier than African hair.

Smoothness

Aligned cuticles lead to smooth and less damaged hair. Indian hair is the least damaged of the three.

Wake Up To Grandma's Advice

Indian samples proved the efficacy of the age-old practice of oiling hair. Vegetable oil is known to penetrate hair, forming a barrier below the cuticle and preventing entry of water into the shaft. It improves hair diameter and lowers loss of vital proteins. Oiling may not directly impact tensile properties, but a favourable change was seen in the break extension value.



■ Compiled by **Damayanti Datta**